AI-Augmented Research

Session 2.2: From Research Question to Verified Finding

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instats Seminar

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Interactive App and Repository

- **Streamlit App URL:**
  Click here to open the Interactive Streamlit App
- **No Installation:** Runs entirely in the browser via Streamlit — no Python or programming knowledge required.
- **Navigation:** Use the sidebar to switch between the six seminar sessions across the three days.
- **Step-by-Step:** Each page walks through worked examples and a Bring Your Own Data (BYOD) interface for that session's core workflows — from research question development and literature scoping to data quality auditing, end-to-end computational pipelines, manuscript integrity checks, and AI disclosure evaluation.
- **Consistency:** Results match the companion Python scripts and prompt library available in the `scripts/` folder of the seminar GitHub repository.
- **GitHub Page URL:** Click here to open the seminar GitHub page

Session 2.2 Overview: One Complete Case Study

Case study: Social media use and academic performance

Research question: What is the relationship between social media use and academic performance among university students?

Workflow: Research question → Data collection → Cleaning → Exploratory analysis → Visualization → Interpretation → Documentation → Verification

Tools: No-code and low-code AI environments throughout. Every step applies the verification disciplines from Session 2.1.

Step 1: Operationalizing the Research Question

Using AI in Tutor mode

1. Ask: “What are the main ways ‘social media use’ has been operationalized in empirical research on academic performance?”
2. Review the response and verify 2–3 cited operationalizations against primary sources
3. Select an operationalization and document the choice and its rationale
4. Ask: “What are the main threats to validity in a cross-sectional survey design for this question?”
5. Evaluate each threat and document the researcher’s response

Verification step: confirm that the operationalization chosen is consistent with at least one peer-reviewed precedent.

Zyphur (2026): operationalization is a judgement task — the researcher decides; AI assists in Tutor mode.

Step 2: Data Collection — Identifying and Accessing Data Sources

Using AI in Search mode

1. Ask: "What publicly available datasets contain measures of social media use and academic performance?"
2. **Verify each dataset mentioned:** does it exist? Is it publicly accessible? Does it contain the variables described?
3. Select a dataset (e.g., a publicly available survey dataset from OSF or Kaggle)
4. Document: source, version, date of access, number of observations, key variables

Verification step: download the dataset and check that the variable descriptions match the AI's description.

This step illustrates the discovery-versus-verification distinction: AI discovers candidate datasets; the researcher verifies each one.

Step 2b: AI-Assisted Information Extraction from Unstructured Sources

Extracting structured information from a PDF codebook

1. Upload the PDF codebook to ChatGPT or Claude
2. Ask: “Extract all items related to social media use and academic performance into a table: item number, item text, response scale”
3. **Verify the extraction:** compare the AI-generated table against the original PDF for 10 randomly selected items
4. Document: extraction prompt, model used, verification result

Verification step: report the error rate in the extraction (number of items with errors / total items extracted).

Gartlehner et al. (2025): AI-assisted extraction accuracy 91.0% — the 9% error rate requires verification.

Step 3: Data Cleaning — AI-Assisted Identification of Issues

Using Julius AI or ChatGPT Code Interpreter

1. Upload the dataset (CSV format)
2. Ask: "Identify missing values, outliers, and inconsistencies in this dataset"
3. Review the AI's report: is each identified issue genuine? Are there issues the AI missed?
4. For each issue, make a documented decision: impute, exclude, or retain with a note
5. Document every cleaning decision and its rationale

Verification step: compare summary statistics before and after cleaning; check that cleaning decisions are consistent with the analysis plan.

Grimmer, Roberts & Stewart (2022): data cleaning decisions are methodological decisions, not technical ones.

Step 4: Data Annotation — AI as a Coder, Not an Authority

Annotating open-ended survey responses

1. Define the annotation scheme in writing: categories, decision rules, examples of ambiguous cases
2. Run the AI annotation on the full dataset
3. Manually code a random sample of 50 responses using the same scheme
4. Calculate Cohen's κ between AI and human coding
5. Report the κ value and the sample size; if $\kappa < 0.7$, revise the scheme and repeat

$\kappa \geq 0.7$ is the minimum acceptable agreement for publication.

Step 5: Exploratory Analysis — Visualizing the Data

Using Julius AI or similar

1. Ask: "Create a histogram of daily social media use hours"
2. **Verify:** does the histogram accurately represent the distribution? Check against summary statistics
3. Ask: "Create a scatter plot of social media use hours vs. GPA"
4. **Verify:** are the axes correctly labeled? Is the scale appropriate? Does the pattern match the correlation coefficient?
5. Ask: "Identify any outliers in the relationship between social media use and GPA"
6. **Verify:** are the identified outliers genuine? Are they data errors or substantively interesting cases?

Verification step: reproduce at least one visualization manually (e.g., in Excel) to confirm the AI's output is correct.

Step 6: Pattern Description — Applying the Sycophancy Check

AI-generated pattern description with sycophancy detection

1. Ask: "Describe the main patterns in the relationship between social media use and GPA in this dataset"
2. Evaluate: is the description accurate? Does it overstate the strength of the relationship? Does it confuse correlation with causation?
3. **Apply sycophancy detection:** ask the AI to identify the weakest aspects of the observed relationship
4. Document the AI's description and the researcher's evaluation

Verification step: check each numerical claim in the description against the underlying statistics.

Zyphur (2026) Competency 5: AI agreement is not evidence; flat agreement is a failure signal.

Step 7: Interpretation — The Researcher's Non-Delegable Role

Interpretation is a **judgement task**. The researcher must:

1. Situate the findings in the existing literature (what do these results add to or contradict?)
2. Assess the alternative explanations (what else could explain this pattern?)
3. Evaluate the limitations (what does this design not allow us to conclude?)
4. Draw the appropriate inference (what can and cannot be concluded from this data?)

AI can assist with steps 1 and 2 (in Search and Validator modes) but **cannot perform steps 3 and 4**. The researcher's intellectual authority is most critical at the interpretation stage.

Zyphur (2026): Dimension 1 — interpretation is a judgement task that must remain human.

Step 8: Documentation — Creating the Reproducibility Record

The reproducibility record

- Document every AI-assisted step: tool, version, date, prompt, output, verification result
- Create a methods section reporting: tools used, prompts used, verification steps, failure-discovery rate at each step
- Save all AI outputs to files with timestamps
- Create a data availability statement: source, version, date of access
- Create a workflow availability statement: tools, prompts, availability for replication

Verification step: ask a colleague to attempt to replicate one step using the documentation provided.
Zyphur (2026) Competency 2: model and parameter specification as a minimum documentation standard.

Step 9: Final Verification — The Structured Failure-Mode Report

Applying Zyphur Competency 6

1. Re-run the key analysis steps and check that outputs are substantively identical (re-run stability test)
2. Verify a random sample of citations in the interpretation section against primary sources
3. Check all numerical claims in the write-up against the underlying data
4. Apply model-heterogeneity: run the key interpretation prompts through a second AI system and compare
5. Document: what was checked, what errors were found, how they were resolved

The structured failure-mode report: Protocol + Threat model + Failure-discovery rate.

No-Code Tools for Computational Research: A Curated Overview

Stage	**Tool**	**Key capability**	**Verification discipline**
Data collection	Octoparse, Browse AI	Web scraping without code	Check against source
Data extraction	ChatGPT, Claude	Extract from PDFs, images	Spot-check against source
Data cleaning	Julius AI, ChatGPT Code Interpreter	Identify and resolve data issues	Compare before/after statistics
Analysis	Julius AI, JASP, Jamovi	Statistical analysis via natural language	Reproduce key statistics manually
Visualization	Julius AI, Tableau Public	Charts and figures	Verify against underlying data
Documentation	Notion AI, Obsidian	Structured note-taking	Review for accuracy

AI Agents for Analytical Tasks: Opportunities and Risks

AI agents — systems that can autonomously execute multi-step analytical tasks — represent the frontier of no-code computational research.

Opportunities:

- Execute complete analytical workflows from a single natural-language prompt
- Dramatically reduce the time cost of routine analytical steps

Risks:

- Each autonomous step introduces a potential error the researcher may not see
- Verification burden increases with agent autonomy
- Reproducibility is particularly sensitive to model updates

Gibney (2025): AI agents are increasingly used for complex, multi-step research processes.

The Human-as-Verifier Discipline Throughout the Case Study

At every step of the case study, the human-as-verifier discipline was applied:

- AI outputs were checked against independent standards
- AI agreement was treated as a potential sycophancy signal
- The researcher's judgement was the element that converted AI-generated material into defensible research findings

This discipline is what distinguishes **AI-augmented research** (valid) from **AI-generated research** (not valid).

Zyphur (2026) Competency 5: human-as-verifier discipline.

Session 2.2 Summary

Three things to carry forward

1. The **end-to-end workflow** (research question → data collection → cleaning → analysis → visualization → interpretation → documentation → verification) is the template for valid no-code computational research
2. The **verification burden does not decrease** with no-code tools; it increases, because the researcher cannot inspect the underlying code
3. The **human-as-verifier discipline** is what distinguishes AI-augmented research from AI-generated research: the researcher's judgement at every step is what makes the findings defensible

References

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Questions and Discussion

Thank you!

Questions?

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